Final Prototype Memo

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Background

Our client is Prof Madero, a computer science professor at Harvey Mudd and a member of the Sustainable Claremont Board. “Sustainable Claremont is an independent nonprofit organization that partners with community organizations, municipalities, businesses, and educational institutions to further environmental health for all” (Sustainable Claremont). She is working with Stuart Wood, the executive director of Sustainable Claremont, and Romeo Lodia, the Community Compost & Climate Resilient Home Program Manager. Sustainable Claremont has many compost piles across the Claremont area at elementary schools. Each compost pile ranges in size up to 4x4x4 ft. Compost piles need to be watered in order to support the microorganisms that facilitate the breaking down of compost. Currently, our clients hand water the piles multiple times a week for about 3 hours using the shower setting on the hose.

Problem Statement

Sustainable Claremont required a reliable and low-labor method for watering its community compost piles. The solution needs to be able to effectively distribute water throughout piles of varying sizes up to 4x4x4 ft, without flushing out the microorganisms needed for decomposition.

Prototypes

The main question our prototypes sought to answer was: how well does our irrigation system maximize water distribution across the entire compost pile while minimizing setup time and effort to use?

Prototype 1: Internal compost water distribution

Figure 1: Water flowing through the PVC pipe and our drip system (left) and PVC pipe in a pile of dirt (right)

The first design alternative we selected to test was a PVC tree design, which would sit inside the compost pile. We originally chose this design because we thought it would give us the best water distribution, our most important objective.

Evaluation plan: The goal of our first test was to test the distribution of water when the pipe is inside the compost. In order to test this, we constructed a model of one of the ‘branches’ of the PVC tree and drilled holes into the pipe 6 inches apart. First, we ran water through the pipe to see if the water came out of each hole evenly, which it did. Then, we grabbed a pile of compost and placed the pipe inside the pile, and ran water through the pipe.

Evaluation results: After running water through the pipe, we found that most of the water seeped through the compost to the bottom instead of permeating throughout the compost pile. We decided that this design alternative would not be as effective at distributing the water evenly as we originally thought. Further, when talking to our client about this issue, we came to the conclusion that designs internal to the compost pile would disrupt the ability to maintain and upkeep the piles.

Prototype 2: Shower head - failed

Figure 2: set up for shower head test (left) and set up for measuring (right).

Evaluation Plan: We planned to see how well the 3D printed shower head would distribute water. We tested this by connecting the showerhead to a PVC pipe with a 3-way connector and putting a hose through the top. We constructed a pile of dirt and mulch, with a glass divider positioned against it, to see how deep the water reached.

Evaluation Result: When we first started testing, it seemed pretty promising. The water distribution was good, and it seemed to be saturating our makeshift compost pile. As testing went on, the hot glue used to secure the center of the showerhead gave out, causing the water to pool together in one big stream. Because of this, we considered this evaluation plan a failure. From this prototype failure, we learned we needed a new way to connect our shower head and shower head cap. We decided to go with a threading design to screw the head together instead of connecting them with glue.

This test directly led to this new head design idea:

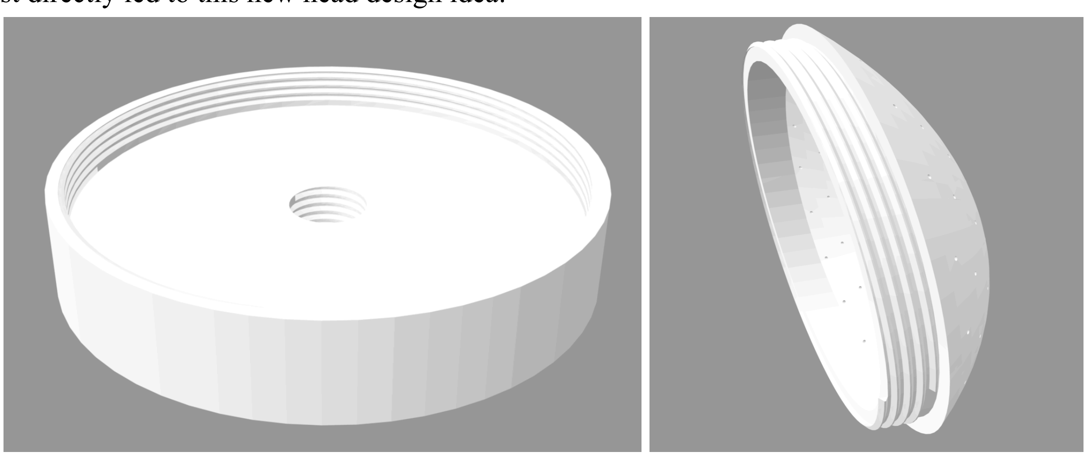

Figure 3: Base of showerhead (left) and cap for showerhead (right).

Prototype 3: Four different heads - prototyping fit

Figure 4: Showerheads varying in thread size, overall size, and hole locations.

For our head prototyping, we went through many iterations of size and thread placement/type.
Evaluation plan: These prototypes were intended to test if the threads of the two components fit together. If they did, we then poured water through them for 5 minutes to see if there was any leakage in the threading connection. After finding no leakage, we connected the showerhead to our frame and moved on to further tests.

Prototype 4: Testing water distribution with new head design (qualitative)

Evaluation plan: We wanted to see the distribution of our prototype with both the shower head as well as PVC water system. This was especially important to test the distribution from the new shower head we made. We set up our prototype over a dry patch of concrete and ran the water system for 5 minutes. We decided to deem the test a success if around 90% of the concrete was wet after we ran the watering system.

Evaluation results: After running the watering system for 5 minutes, we saw that 100% of the concrete that was in the 4x4 square was wet from our watering system, leading to our prototype test being a success. During this test, however, we realized that using PLA filament for the shower head could cause it to break down over time and not be durable. This led us to look into the use of other materials for the shower head. We settled on using resin for our shower head.

Final Resin Head Design

This resin head was the final design we settled on after a successful water test with the firmament version of this head. The resin is much more durable and will not have any issues under the pressure of the water for a long time. The resin has a threaded screw on capability, which keeps it water sealed and able to perform its function without any long term issues.

Prototype 5: Resin head: water distribution with cups, (quantitative)

Test 1									
Quad 1 (Green)	Water (cm)	Quad 2 (Blue)	Water (cm)	Quad 3 (Orange)	Water (cm)	Quad 4 (Purple)	Water (cm)	Middle Quad (Green)	Water (cm)
cup 1	0	cup 1	0	cup 1	3.2	cup 1	0.6	cup 1	3.9
cup 2	3.3	cup 2	0	cup 2	0.1	cup 2	2.3	cup 2	6.8
cup 3	0.1	cup 3	0	cup 3	0	cup 3	2.2	cup 3	6
cup 4	8.4	cup 4	0.2	cup 4	0	cup 4	0.3	cup 4	9.6
cup 5	0	cup 5	6.4	cup 5	2.6	cup 5	2.4	cup 5	5.5
Averages (cm)	2.36		1.32		1.18		1.56		6.36
Test 2									
Quad 1 (Green)	Water (cm)	Quad 2 (Blue)	Water (cm)	Quad 3 (Orange)	Water (cm)	Quad 4 (Purple)	Water (cm)	Middle Quad (Green)	Water (cm)
cup 1	0	cup 1	0	cup 1	0.2	cup 1	0.2	cup 1	5
cup 2	0.2	cup 2	0	cup 2	1.4	cup 2	5.5	cup 2	6.8
cup 3	0.1	cup 3	0	cup 3	0.1	cup 3	2.6	cup 3	7.2
cup 4	0	cup 4	8.2	cup 4	4.8	cup 4	3	cup 4	5.7
cup 5	2.6	cup 5	7.6	cup 5	1	cup 5	0	cup 5	10.5
Averages (cm)	0.58		3.16		1.5		2.26		7.04
Test 3									
Quad 1 (Green)	Water (cm)	Quad 2 (Blue)	Water (cm)	Quad 3 (Orange)	Water (cm)	Quad 4 (Purple)	Water (cm)	Middle Quad (Green)	Water (cm)
cup 1	6.5	cup 1	0	cup 1	5	cup 1	0	cup 1	5.6
cup 2	0	cup 2	0	cup 2	0	cup 2	0	cup 2	2.5
cup 3	0	cup 3	0	cup 3	0	cup 3	5.5	cup 3	8.6
cup 4	0	cup 4	0.5	cup 4	0	cup 4	5.5	cup 4	1
cup 5	8	cup 5	7	cup 5	3.6	cup 5	0.1	cup 5	6
Averages (cm)	2.9		1.5		1.72		2.22		4.74
	Quad 1		Quad 2		Quad 3		Quad 4		Middle Quad
Overall Average (cm)	1.946666667		1.993333333		1.466666667		2.013333333		6.046666667

For prototype 5, we wanted to test how the overall distribution was for our irrigation system. We defined distribution as being an equal amount of water received in 4 different quadrants under our system. For our test to be successful, we wanted 4 corner quadrants (Figure 5; top left green, blue, pink, orange cup locations) over 3 different tests, to be within 30% or more of each other in cms of water filled. In the middle quadrant (figure 5: middle green cups), we want them to have at least 2x as much water, since the pile in the middle has the most compost. The steps we followed to run this test were to,

1: Place the prototype on a flat surface and attach the three hoses to a water source.
2: Place a black taped cross that evenly breaks the bottom of the structure into even quadrants.
3: Place five empty cups in a plus shape (figure 5) in each of the four quadrants' centers, and then place 5 empty cups in a plus shape in the intersection zone of the black lines.
4: Turn on the hoses, and once water starts to come out of the system, start a two-minute timer and let the system run for those two minutes.
5: After two minutes, stop the hose and wait for all the water to stop dripping out of the prototype.
6: Measure with a ruler how much water in cm has filled up each cup, and take the average amount of water in each group of 5 cups for the 5 different groups (quadrants) of cups.
7: Repeat steps 1-7 for three different trials and record the means of each quadrant's water height.
8: Find the mean of the three different trials for each quadrant to get an overall average water received per quadrant.
9: Finally, compare how similar each of the 4 quadrants was and how much more water the middle or 5th quadrant was in comparison.

What we learned:
After the three trials, we recorded that on average, quadrant 1 received 1.94 cm of water, quadrant 2 received 1.99 cm of water, quadrant 3 received 1.46 cm of water, quadrant 4 received 2.01 cm of water, and finally, the center or middle quad received 6.04 cm of water.

When we did our data results:
1.94 vs 1.99: 1.94/1.99 = 0.974 → within 10%
1.94 vs 2.01: 1.94/2.01 = 0.965 → within 10%
1.99 vs 2.01: 1.99/2.01 = 0.990 → within 10%
1.46 vs 1.94: 1.46/1.94 = 0.752 → NOT within 10% → within 30%
1.46 vs 1.99: 1.46/1.99 = 0.734 → NOT within 10% → within 30%
1.46 vs 2.01: 1.46/2.01 = 0.726 → NOT within 10% → within 30%
6.04 is 2x greater than [1.94, 1.99, 1.46, 2.01]

This means that three of the four quadrants were similar to a 10% value, but only the purple cup quadrant was not in a 10% similarity rate. The middle quadrant was above 2x all other quadrants, so that was a success. This means overall that the 3/4 cup groups were evenly distributed, and the center group received 2x or more than the outer groups. The last quadrant was not distributed evenly in the 10% range, but was even in the 30% range. There may have been some uncertainty in our measurements as a result of inconsistent measuring methods (using a ruler and estimating to the nearest mm) and variations in the exact amount of time the water was running for. Overall, we still believe the test showed the general trend that we wanted, more water in the middle and pretty even distribution of water around each side. After this test, and from our results, we realized our resin head had a very small leakage near the connection thread of the cap and grate, which was causing that different water pattern to occur, messing up the distribution of water on one side (Orange cup side). We used a resin epoxy hardener to fix this leakage, which created the overall even distribution on all sides we wanted. We then, with our hardened shower head, felt confident going to the clients to test our prototype at their compost sites.

Prototype 6: Structure 1 - tested weight

Evaluation plan: We decided to test the durability of our prototype since water would be flowing through the top part of the structure, and it would need to be able to support it. For the structure to be deemed durable, we decided that the most it could move when weights were added was 3 cm. The steps we followed to run this test were as follows:

1. Place the structure next to a wall and put a piece of tape on the wall at the top of the structure.
2. Place 5 lbs on both sides of the structure for even distribution, as shown in the photo above.
3. Measure the distance between the bottom of the piece of tape and the top of the structure.
4. Add 5 lbs on both sides of the structure.
5. Repeat steps 3 and 4 until reaching 40 lbs.

Weight (lbs)	Displacement (cm)
5	0.5
10	1.125
15	1.75
20	2.5
25	3.1
30	3.875
35	4.75
40	5.5

From this test, we learned that our structure is not able to hold a lot of weight. While the structure did not break, it had a significant weakness and was not able to hold up even light weights. This led us to believe that over time, with the weight of the water going through the structure, it would eventually wear down and break. Since we want our structure to be durable, this prototype needed to be changed. We decided to add extra support to the sides.

Prototype 7: Structure 2: added middle connectors and tested weight

Evaluation plan: After adding the additional supports to our structure, we wanted to make sure that the supports were able to help the durability of the structure. We decided that the test would be a success if it was able to have only 3 cm of displacement when 40 lbs were added. To test this, we did the following:

1. Place the structure next to a wall and put a piece of tape on the wall at the top of the structure.
2. Place 5 lbs on both sides of the structure for even distribution, as shown in the photo above.
3. Measure the distance between the bottom of the piece of tape and the top of the structure.
4. Add 5 lbs on both sides of the structure.
5. Repeat steps 3 and 4 until reaching 40 lbs.

Weight (lbs)	Displacement (cm)
5	0
10	0
15	0
20	0
25	0
30	0
35	0
40	0

From this test, we were able to see that the additional supports that we added to the structure were a success since there was no displacement for all weights. This provided us with confident information that our structure would not wear down and break from the constant weight of water through the system, which is significantly lighter than 40 lbs.

Prototype 8: Final prototype


We have worked on designing our prototype by solving the question: How does the shower head irrigation system maximize water distribution across the entire compost pile while minimizing setup time and effort to use?

To test this question, we took our prototype out to one of the compost sites with our clients. The client told us that we would count it as a success if, after 10 minutes of watering, an internal part of the compost could be squeezed and one drop of water would fall out. We tested this by doing the following:

1. Set up the structure around a compost pile.
2. Turn on the watering system and leave it on for 10 minutes.
3. Reach into the center of the compost pile and grab a handful of dirt.
4. Squeeze the pile of dirt in hand and observe the number of drops of water.
5. Repeat step 3 & 4 with a different location in the pile of compost.

The client tested our prototype and was satisfied with the internal distribution of the water compost pile. Both times she grabbed a handful of dirt and squeezed, one drop of water fell out. This result validates our design choices throughout this process because it shows that we were able to achieve our main objective of maximizing water distribution. This question aligns with our first two objectives: maximizing water distribution while minimizing effort to use. We have created a valid prototype that works, aligns with our functions and constraints, and can be used by the clients today!

Impact of Design: Our design allows our client to water their compost piles hands-free. Our design allows them to turn on the hose, wait a desired amount of time, and turn it off. This is a hands-free design that evenly waters their piles the same way every time. The client has to put the product over the pile once, and then they never have to touch it again, as the open structure allows for work to be done on the pile without ever moving the structure. Our clients can now water their piles more efficiently and with less work!

Constraints

1.) Must operate autonomously once turned on: Once the water is turned on, it doesn't need to be touched until it is turned off; therefore, our prototype meets this constraint.
2.) Must be safe for children to be around: The pile doesn't have any sharp edges, isn't made of toxic materials, and is clearly visible. Our clients noted that the frame also provides an opportunity to put educational posters on, which makes it not only safe for children, but also educational.
3.) Must not use water pressure higher than 80 PSI to preserve microorganisms: The water pressure can be adjusted through the hose splitter knobs, and when we tested our prototype on the compost pile, the clients noted that the water pressure was low enough to not harm the microorganisms.
4.) Must be compatible with every range of pile up to 4x4x4ft compost pile sizes: Our design was tested on a pile at the largest they normally get, and spread water throughout the pile. Furthermore, the design can be modified by removing parts and cutting the PVC to fit piles of smaller sizes.
5.) Must withstand rain, sun, and wind for 24 hours a day: The materials used can withstand the elements (discussed more in the materials section).
6.) Must be compatible with existing water sources: Our prototype screws onto a standard hose/spigot, allowing it to be used anywhere with a hose.
7.) Must cost less than $150 to design: Our bill of materials is about $200; however, much of the cost is through the amount of PVC used. Smaller compost piles won't require as much PVC, which would cut down on the cost.
8.) Must withstand water pressure without breaking: When testing our design, it withstands the water pressure of the hose throughout the typical watering duration.

Objectives

Our main objectives for the prototype are as follows:

1.) Maximize water distribution: Our metric for testing the distribution is both qualitative and quantitative. Our qualitative objective is the client's satisfaction with the watering of the compost pile. When we tested our prototype on site with the clients, they grabbed a handful of compost from the pile and squeezed it. One water came out of the handful, meaning the compost had been sufficiently watered. Further, from our cup water distribution test, we saw success in water distribution, giving us confidence that our design maximizes water distribution.
2.) Minimize set-up time: The prototype has a single one-time 20-minute assembly time, and after that, the user just has to turn on the water, and the prototype never needs to be touched again.
3.) Maximize transportability: The prototype can be disassembled very easily and takes less than 20 minutes to disassemble into a transportable size. After testing our prototype at the compost site, we quickly disassembled it and transported it back to school in Chase's car.
4.) Minimize deformation of compost pile: The prototype sits around the perimeter of the pile, not inside or on top of the pile. This means that it doesn't ever touch the pile. Furthermore, when testing the watering device, the pile didn't have any divots from the watering, showing us that the pile wasn't deformed in any way.
5.) Minimize maintenance requirements: All materials in the design are durable and should require little maintenance. We haven't had a chance to test our design's durability over a long time period, but we are confident that if something were to break, it would be a relatively easy fix.

6.) Minimize water usage: After testing our design, the clients said that using our design would allow them to water the pile at a lower water pressure and for a shorter time, which would reduce the water usage from their current solution.
7.) Minimize cost: We minimized the cost through our material choice while still maintaining a durable design. This will help minimize the cost down the line as our design will be durable.

Functions

Distribute water evenly to the compost pile

- Even water distribution provides every part of the compost pile with the moisture needed for decomposition to occur.

Resist gravitational forces

- The device must remain stable and resist collapsing under its own weight so it can be used continuously without any adjustments.

Connect to a water source

- The device system must easily attach to the client's water supply to ensure it has access to water without requiring any extra effort.

Regulate water flow rate

- Controlling the water pressure lets the client adjust the moisture levels of the compost according to the pile's needs.

Our design passed our water evenness distribution test, so the function of "Distribute water evenly to the compost pile" was successfully met, and the clients at the compost pile were very happy with the results. Our structure after our strength test showed it can not only resist gravity, but also 40+ lbs of weight on top, so we can confidently say our structure will resist gravity and continue to stand up under the elements and time. Our prototype does allow for a connection to a water source by connectors attached to the structure for hoses. Finally, in our final prototype, the hose is connected to the hose splitter that splits the hose into 3 separate parts going to our prototype. The hose splitter has a knob allowing the water pressure to be adjusted for each section of the prototype. If the user wants to adjust the water pressure going to the head, they can simply adjust that knob to their intended water pressure.

Material Choices

Our final prototype is made up of a PVC frame and connectors, with a showerhead that is 3D printed from resin and 2 PVC connectors printed from PLA, with a plastic hose splitter.

PVC Frame: The material choice of PVC is supported when we look at our constraints, objectives, and functions.

- Constraints
    - Safe for kids: Our PVC frame doesn't have any sharp edges or other dangerous materials that make the frame safe for children to be around.
    - Fit piles of different sizes: PVC is also modular and can be cut to specific sizes and connected in different configurations, allowing our design to be modified to fit compost piles of different sizes

    - Environmental conditions: PVC is durable and waterproof and has a lifespan of 50-80 years outdoors, according to. How Long Will PVC Pipe Last in the Sun. This means that we can leave our design outside, year-round, without worrying about environmental degradation.
    - Connect to water source: Our PVC pipe watering system has hose connectors that allow it to connect to an existing hose/water source
- Objectives
    - Water distribution: Since we built our frame out of PVC, we can also run water throughout our frame, allowing us to maximize the distribution of water throughout the pile
    - Set up time: PVC can be easily assembled and disassembled at the compost site.
    - Cost: At about $15/20ft of PVC, PVC pipe provides a relatively low-cost solution for our frame.
- Functions
    - Resist gravitational forces: Our PVC frame is self-supporting
    - Connect to water source: Our PVC pipe watering system has hose connectors that allow it to connect to an existing hose/water source

Resin Shower Head: Originally prototyped as a PLA showerhead, our choice of resin is supported by our objectives, constraints, and functions.

- Constraints
    - Environmental conditions: According to Is Resin Truly Weatherproof for Outdoor Use?, most types of resin, especially epoxy and polyester resins, are weatherproof and can withstand exposure to rain, sunlight, and temperature fluctuations without significant degradation.
    - Cost: Printing our showerhead costs about $19 and is generally a cheap alternative
    - Water pressure: Resin is better able to withstand the hose pressure compared to our PLA prototype, which burst in a test.
- Objectives
    - Set up time: Choosing 3D printed resin allowed us to easily create a thread, allowing the head to easily be screwed onto the rest of the frame.
- Functions
    - Water distribution: 3D printed resin allowed us to easily manufacture a curved face into our showerhead, which allowed the showerhead to spread the water throughout the pile.

3D printed PLA connectors: We decided to 3D print our PVC connectors to allow us to manufacture them to a tight tolerance in a short amount of time, allowing us to slip fit our PVC pipes into the connectors. We chose to use PLA because,

- PLA is cheaper to print than resin
    - Makerspace charges us for using the resin printer
- PLA takes less time to print than resin
    - We were in a time crunch and needed to get our project done; we didn't have time to print a resin connector and have it not work.
- While not the strongest material, PLA was enough to support weight in our weight test.